

New Physical Analysis of 10-Gb/s Transparent Optical Networks

D. Penninckx and C. Perret

Abstract—We give a new way to consider physical impairments in transparent networks, including the Kerr effect. Extensive numerical simulations show the tradeoff between the network engineering rule tightness and the required physical margins for light-path establishment.

Index Terms—Networks, optical communication, optical fiber networks, optical switches.

I. INTRODUCTION

IN OPAQUE networks, the signal is regenerated at each node. The optical connections are point-to-point and their design can be tailored to the transmission. On the contrary, transparent networks avoid regeneration at each node, reducing the hardware cost. However, it is impossible to consider *a priori* all optical connections. Physical impairments such as noise, linear, and nonlinear (NL) propagation distortions will cumulate along the lightpaths. Up to now, experiments have shown that transparent networks are technically feasible since long distances could be reached and transparent optical nodes could be cascaded [1]–[3]. Wavelength continuity, protection, and restoration have been considered. However, the problem of knowing if a connection will fail or not because of physical impairments before its establishment has only been partially addressed. “Simple” cases such as polarization-mode dispersion (PMD) alone [4], or four-wave mixing (FWM) alone [5] were considered. The general problem is much more complex, especially when dealing with interplay between the Kerr effect and chromatic dispersion. At least two questions have to be answered: Which simple physical criteria should be used to make the decision? What is the physical design (the engineering rules) of the network? An engineering rule may be, for instance, a constraint on the amplifier spacing or on the dispersion map. Only a statistical analysis is able to give an answer.

For this purpose, we have numerically simulated a random walk in a transparent network. The aim is to find a relation between the bit-error rate (BER) of any connection and its physical parameters. As a matter of fact, if all connection parameters (dispersion map, fiber features, amplifier features, optical powers,...) were known, the BER would be a deterministic function of these parameters. However, this function is much too complex, impossible to compute in a reasonable duration and the amount of information it requires to flood in the network is much too large. All this makes such an

approach hardly realistic in static transparent networks and impossible in dynamic networks for which connections have to be established in a few seconds or less. Hence, only a few “cumulative” criteria have to be found. By “cumulative,” we mean that the criteria may be updated after each amplifier span of the connection without losing information. For instance, the residual dispersion is a cumulative parameter but not the optical power: the whole power distribution along the line has to be known. The feature “cumulative” allows us to reduce the set of criteria. The criteria have to be carefully chosen in order to know the BER with sufficient accuracy. Note that engineering rules may simplify this study. For instance, in the extreme case of a perfectly homogeneous network (for which only one type of fiber is used, the launched optical power is controlled to a constant value, the amplifier span length is constant, and so on,...) the number of amplifier hops and a reference table would be sufficient to determine whether a connection is physically feasible or not [6]. But future transparent networks cannot be completely homogeneous.

In the following, we will explain why and which cumulative criteria we chose, how we numerically simulated a random walk in transparent networks, and how an important engineering rule that has been proposed in the literature [1], [3] may impact the results.

II. CHOICE OF CUMULATIVE PARAMETERS

Excepting transient effects due to the amplifier behavior, the physical phenomena involved in transmission are chromatic dispersion, Kerr effect (divided into self-phase modulation (SPM), cross-phase modulation (XPM), and FWM), Brillouin scattering (SBS), Raman scattering (SRS), optical signal-to-noise ratio (OSNR), crosstalk, PMD, polarization-dependent loss and gain (PDL/PDG), and filter concatenation. OSNR and the residual chromatic dispersion D_{res} are two obvious cumulative criteria. Keeping in mind that a connection, while established, should remain stable, PMD, which is a statistical process, should be considered as a worst case only. The PMD coefficient is one cumulative criterion. PDL and PDG should remain sufficiently low to be negligible in the whole network through component specification (engineering rule). Since PMD is also low in the transparent path, there is no interaction between PMD and PDL/PDG. Moreover, the establishment of a new connection must not disturb the already established connections, so that we have to avoid any significant FWM. Hence, one engineering rule is to avoid the use of dispersion-shifted fiber in the *C*-band, which is not too constraining in most countries. Similarly, crosstalk should

Manuscript received November 26, 2002; revised January 4, 2003.

The authors are with Alcatel R&I, Marcoussis 91460, France (e-mail: Denis.Penninckx@alcatel.fr).

Digital Object Identifier 10.1109/LPT.2003.809977

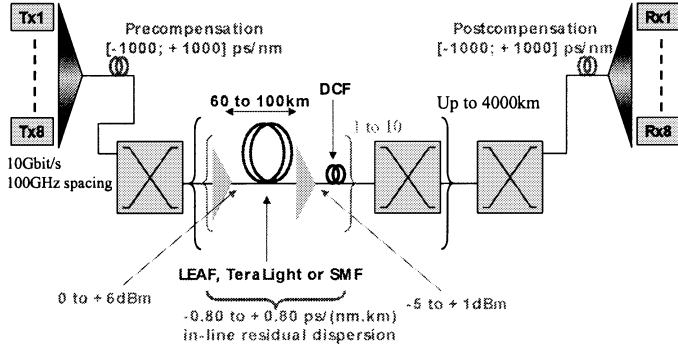


Fig. 1. Representation of the simulated link.

also be avoided through engineering rules, which becomes feasible [2]. SBS is easily negligible since the channel power is usually below SBS threshold. However, SPM and XPM and their interplay with chromatic dispersion cannot be avoided in practice through engineering rules. Hence, cumulative criteria have to be found to assess the amount of SPM and XPM. For this purpose, we have chosen to use the NL phase, which is the average phase variation due to Kerr effect [7]

$$\phi_{NL} = \frac{2\pi}{\lambda} \int_0^L \frac{n_2}{A_{eff}(z)} P(z) dz \quad (1)$$

where λ is the wavelength, n_2 the NL coefficient, $P(z)$ the considered channel optical power along the fibers, and $A_{eff}(z)$ the effective area of the fibers.

SRS mainly induces a power variation of the channels which can be taken into account through Φ_{NL} and OSNR. The amplifier transients may also be taken into account in power margin but a separate study is required. Filter concatenation may also distort the signal but proper filter specification or use of wavebands [1] make its impact negligible.

Hence, we have chosen four cumulative criteria, that we call quality of transmission vector: OSNR, the PMD coefficient, D_{res} , and Φ_{NL} . To make PMD negligible, the PMD coefficient should be below a given value (usually 10%–15% of the bit period) [4]. We now need to find the function linking BER to OSNR, D_{res} , and Φ_{NL} . With a good approximation, BER is given by [8]

$$BER = \frac{1}{2} \cdot \text{erfc} \left(\frac{Q}{\sqrt{2}} \right) \quad \text{with } Q = Q' \cdot \sqrt{OSNR} \cdot \sqrt{\frac{B_{ref}}{B_e}} \quad (2)$$

B_{ref} is the reference bandwidth of the OSNR (usually 0.1 nm or 12.5 GHz) and B_e is the electrical bandwidth of the receiver, around 7.5 GHz at 10 Gb/s with forward-error correction. Q' usually ranges from 0% to 75% for nonreturn-to-zero (NRZ).

Thus, the aim is to find the function f linking Q' to D_{res} , and Φ_{NL} . We now present the numerical simulations of a random walk in a 10-Gb/s transparent network to solve this problem.

III. RANDOM WALK IN 10-Gb/s TRANSPARENT NETWORKS

All the parameters of the random walk are randomly drawn to avoid redundancy and to sweep over a panel of combinations as large as possible (Fig. 1). Nevertheless, the variation ranges

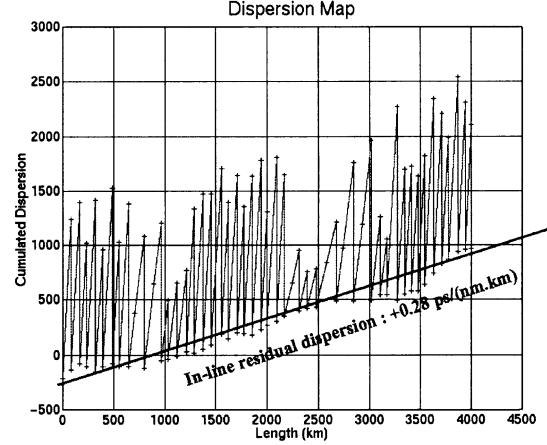
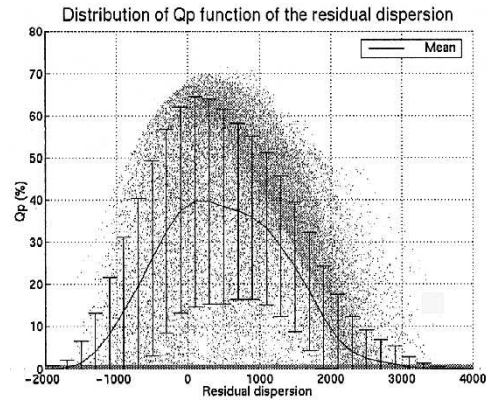


Fig. 2. Example of dispersion map.


 Fig. 3. $Q' = f(D_{res})$.

of the parameters follow loose engineering rules, summarized in Fig. 1. One example of dispersion map is given in Fig. 2.

The propagation of eight 100-GHz spaced NRZ channels is simulated, which is sufficient to assess the influence of XPM. Nodes are separated from one to ten amplifier spans of the same fiber which can be either LEAF, TeraLight, or SMF. However, the fiber type changes from link to link with a probability of 1/3. The fiber features are generated randomly around mean values. The amplifier spacing is randomly drawn between 60–100 km. The optical power per channel is drawn between 0–6 dBm in the line fiber and 5 dB less in the dispersion compensation fiber (DCF). Precompensation and postcompensation are randomly drawn between -1000 and $+1000$ ps/nm and the length of DCF is calculated according to an in-line residual dispersion randomly drawn between -0.8 and $+0.8$ ps/(nm · km) taking into account 5% inaccuracy in its length. Q' is computed after each amplifier span.

IV. RESULTS IN THE GENERAL CASE

Fig. 3 represents Q' as a function of D_{res} for 60 000 simulations obtained with the assumptions given above. The mean value and the standard deviations of Q' are also plotted. Both indicate the relevance of the choice of the engineering rules but the mean value gives a clue of the performance while the standard deviation allows to assess the confidence in the estimated performance. Clearly, because of such high standard deviations,

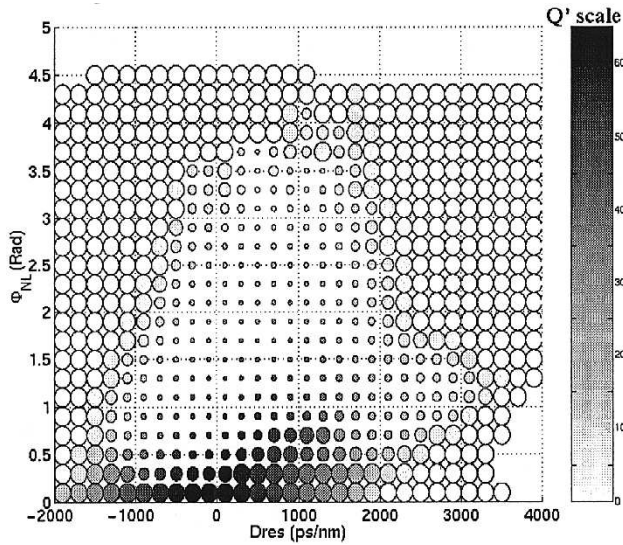


Fig. 4. $Q' = f(D_{\text{res}}, \Phi_{\text{NL}})$ in the general case.

due nonlinearities, a single linear parameter (D_{res}) is not enough to determine with certainty the value of Q' .

Fig. 4 also shows mean values and standard deviations of Q' but as a function of both D_{res} and Φ_{NL} . The gray scale gives the mean value on a delimited area; the darker the disk, the higher Q' . The disk size indicates the accuracy; the wider the disk, the higher the accuracy. The largest disks correspond to 5% standard deviation or below. Note that the accuracy is mainly interesting when Q' is high (above 30%). For Φ_{NL} below 0.5 rad, the result is sufficiently accurate to know whether or not a connection is possible. But 0.5 rad only corresponds to around 700 km. Hence, no longer connection would be guaranteed to be feasible with sufficient confidence, annulling the interest of transparency. At least one additional engineering rule is required to increase the accuracy.

V. RESULTS WITH ONE ADDITIONAL ENGINEERING RULE

In [1], a new simple engineering rule confirmed in [3] has been proposed for transparent networks: choosing the DCF length in order to have constant precompensation, postcompensation, and in-line residual dispersion, whatever the channel and the node. We have added this engineering rule in our simulations, taking unavoidable technological margins (Fig. 5). As an example, we assume that precompensation, postcompensation, and in-line residual dispersion were respectively -200 ± 20 ps/nm, $+400 \pm 40$ ps/nm, and 0.28 ± 0.14 ps/(nm · km) [1]. As expected, 1) the mean value of Q' is higher and, hence, allows us to reach longer distance and 2) the accuracy becomes sufficient to be confident of the value of Q' up to around 2000 km (Φ_{NL} below 1.5 rad). Accuracy is the key point which allows us to avoid 1) trying to establish connections that are not possible for physical reasons, and conversely, 2) not considering actually possible optical paths.

Using a polynomial interpolation of the values of Q' , we easily get an analytical expression of f . Hence, knowing OSNR, D_{res} , and Φ_{NL} in each link for each channel, OSNR, D_{res} , and

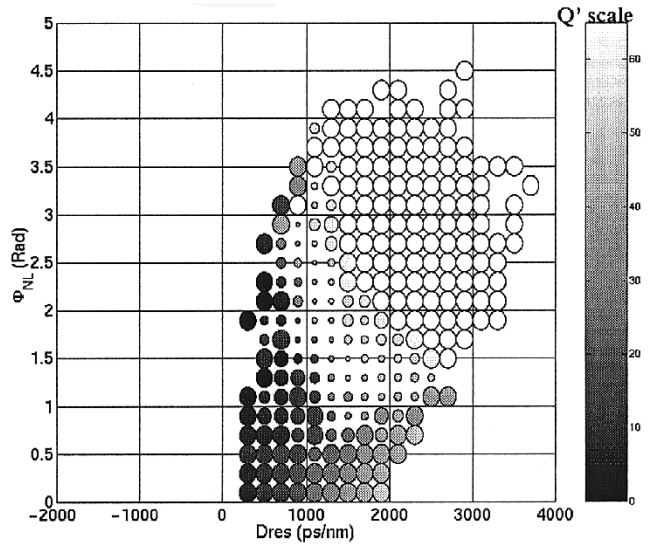


Fig. 5. $Q' = f(D_{\text{res}}, \Phi_{\text{NL}})$ with constant dispersion map.

Φ_{NL} can be computed for any required optical connection and the BER is known accurately.

VI. CONCLUSION

We have analyzed the physical impairments in transparent networks, including the Kerr effect in a new way. We have shown that few cumulative criteria are required to know if connections are physically feasible before establishment. A tradeoff is necessary between the tightness of the engineering rules to design the network and the accuracy of the information given by the criteria. Through numerical simulations, we have shown that using only four criteria and an appropriate dispersion map allows us to know the BER with sufficient accuracy up to 2000 km.

REFERENCES

- [1] D. Penninckx, G. Charlet, J.-C. Antona, and L. Noirie, "Experimental validation of a transparent waveband-based optical backbone network," in *Proc. ECOC'02*, Copenhagen, Denmark, Sept. 2002, Paper 6.4.4.
- [2] I. Tomkos *et al.*, "Ultra-long-haul DWDM network with 320×320 wavelength-port broadcast & select OXCs," in *Proc. ECOC'02*, Copenhagen, Denmark, Sept. 2002, Postdeadline Paper PD2.1.
- [3] I. Tomkos *et al.*, "Dispersion map design for 10 Gb/s ultra-long haul DWDM transparent optical networks," in *Proc. OECC'02*, Yokohama, Japan, July 2002, Postdeadline Paper PD1.
- [4] M. Ali, D. Elie-dit-Cosaque, and L. Tancevski, "Network optimization with transmission impairments-based routing," in *Proc. ECOC'01*, Amsterdam, Netherlands, Sept. 2001, pp. 42–43.
- [5] M. Ali, V. Leboucher, and D. Penninckx, "Intelligent lightpath selection scheme," in *Proc. ECOC'02*, Copenhagen, Denmark, Sept. 2002, Paper S.2.07.
- [6] A. Ehrhardt, N. Hanik, A. Gladisch, and F. Rumpf, "Field demonstration of a transparent optical 10 Gbit/s-WDM-network based on normalized transmission sections," in *Proc. OFC'02*, Anaheim, CA, Mar. 2002, Paper TuH2.
- [7] Y. Frignac and S. Bigo, "Numerical optimization of residual dispersion in dispersion-managed systems at 40 Gbit/s," in *Proc. OFC'00*, Baltimore, MD, Feb. 2000, Paper TuD3.
- [8] D. Penninckx and O. Audouin, "Optically preamplified systems: Defining a new eye aperture," in *Proc. OFC'98*, San Jose, CA, Feb. 1998, Paper WM36.